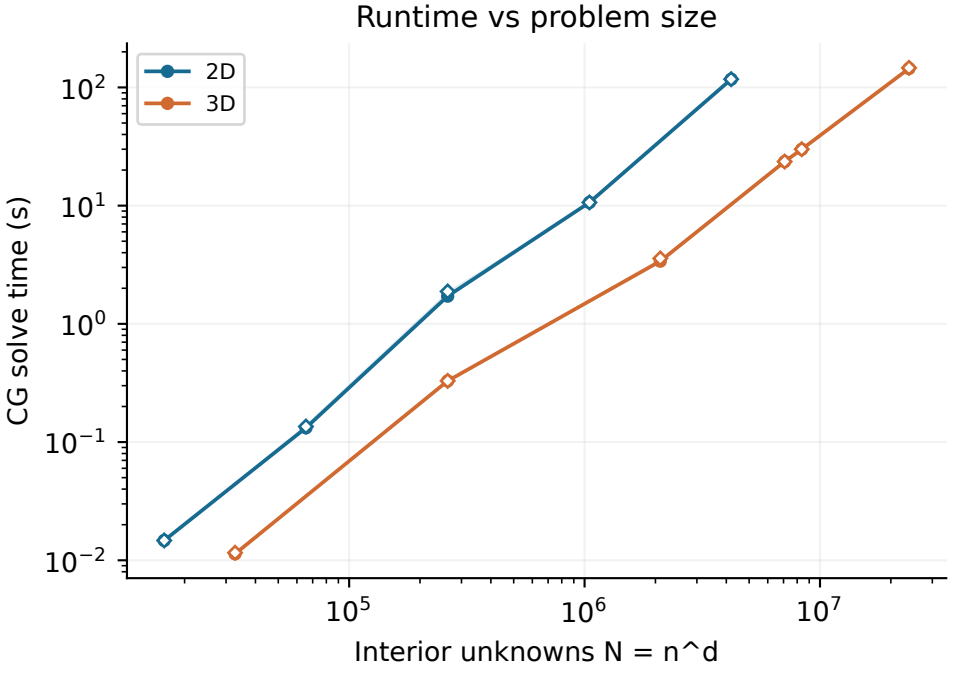
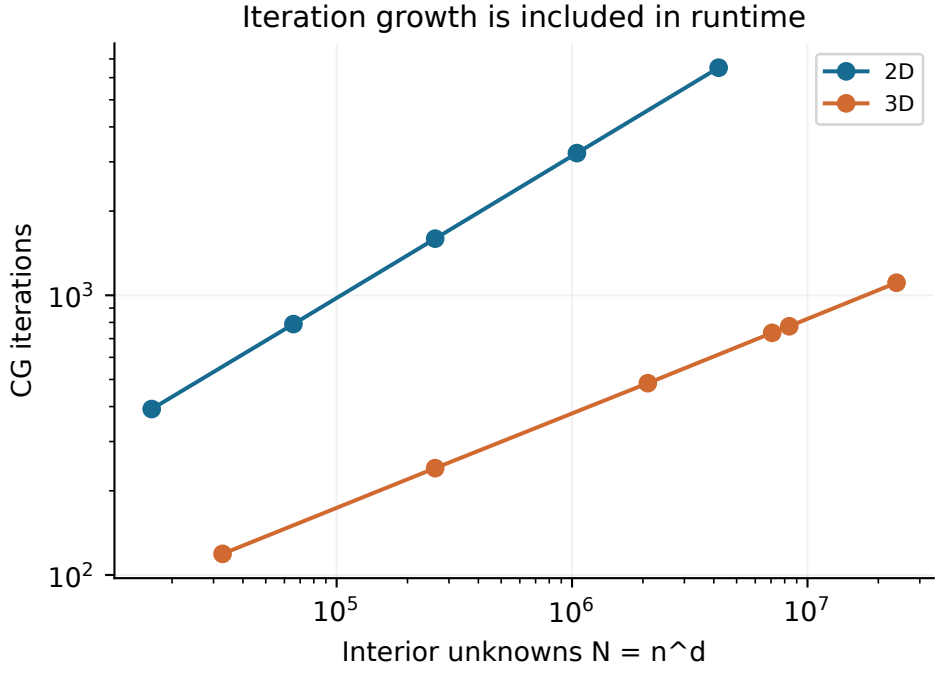
# Serial scaling: runtime and iteration count

Vanda | audited CPU measurements | 3 independent runs per configuration

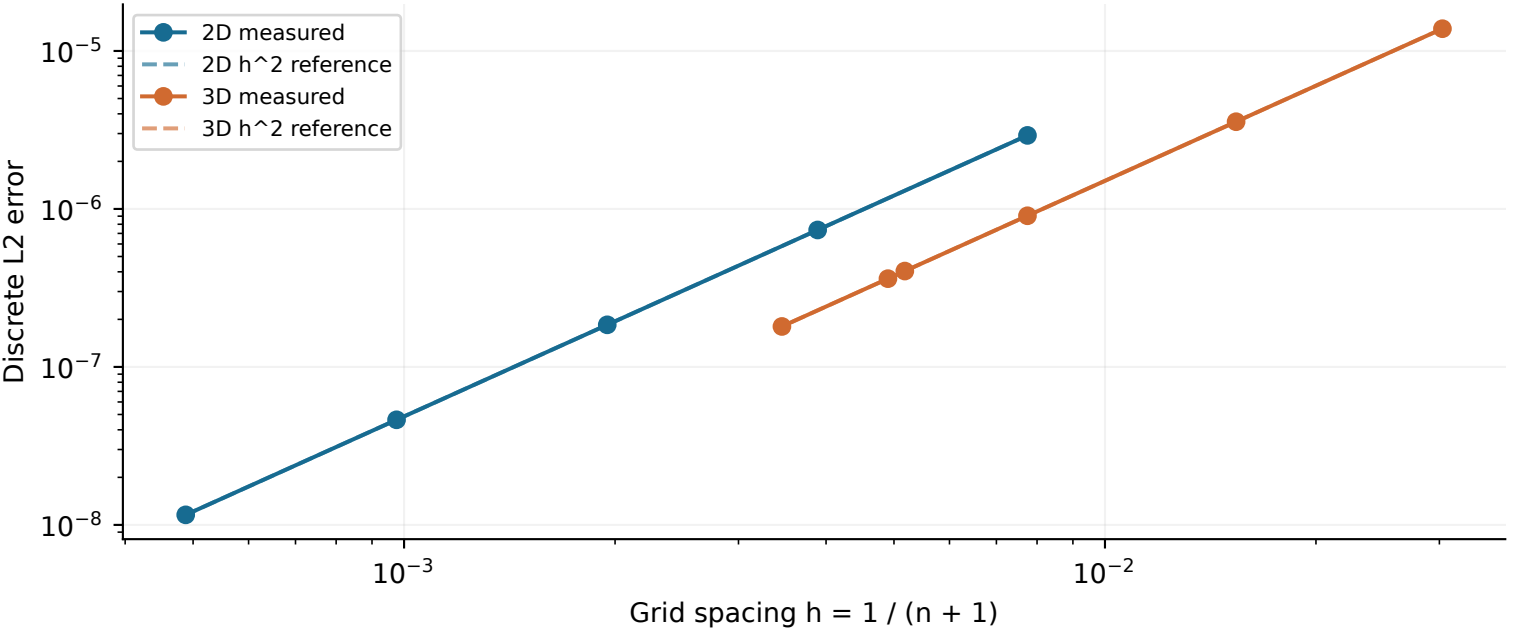



Line/filled marker: minimum; hollow diamond: median; shading: observed min-max (3 runs, not a CI).
The n=203 serial reference is from the weak-scaling supplement; source allocations are retained in the tables.

# Validation: serial discretization error

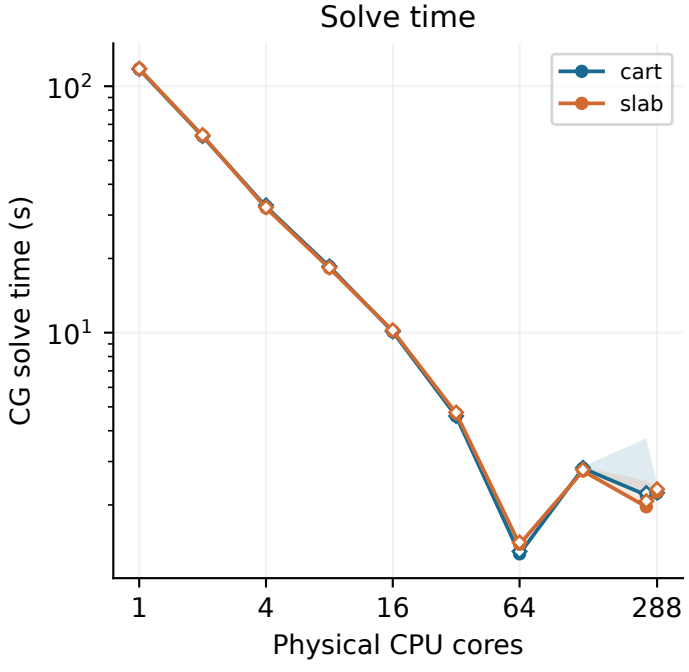
Vanda | audited CPU measurements | 3 independent runs per configuration

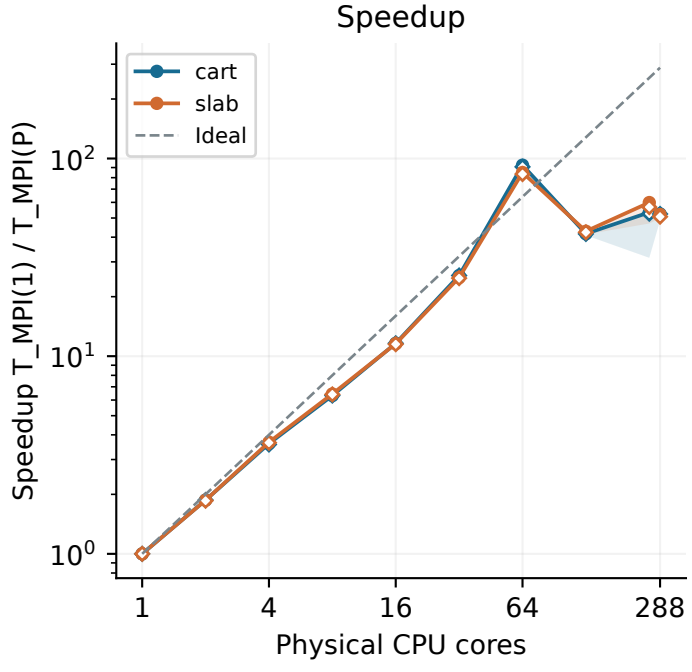


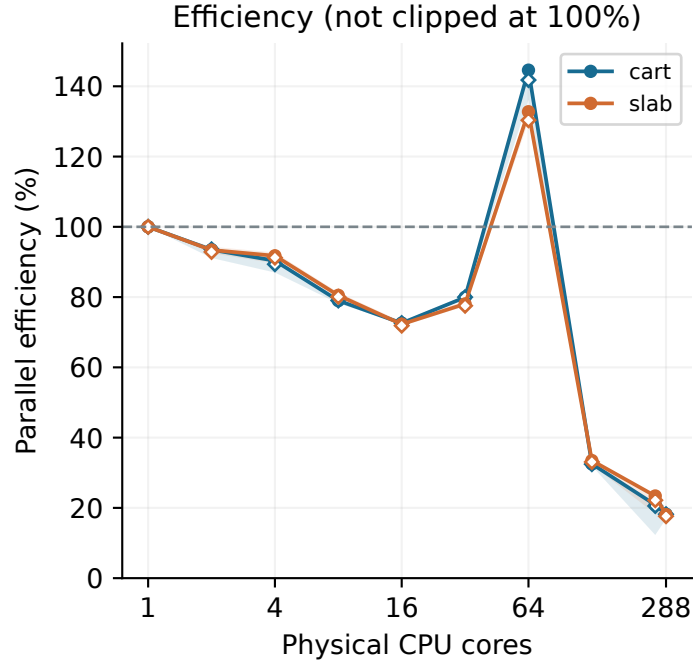

Dashed lines show h^2 trends anchored at each dimension's coarsest mesh; they are not fitted models.
Every retained parallel sample also passes the same-mesh serial iteration/error check and true residual < 1e-8.

# 2D strong scaling | n=2048 | through 288 physical cores

Vanda | audited CPU measurements | 3 independent runs per configuration



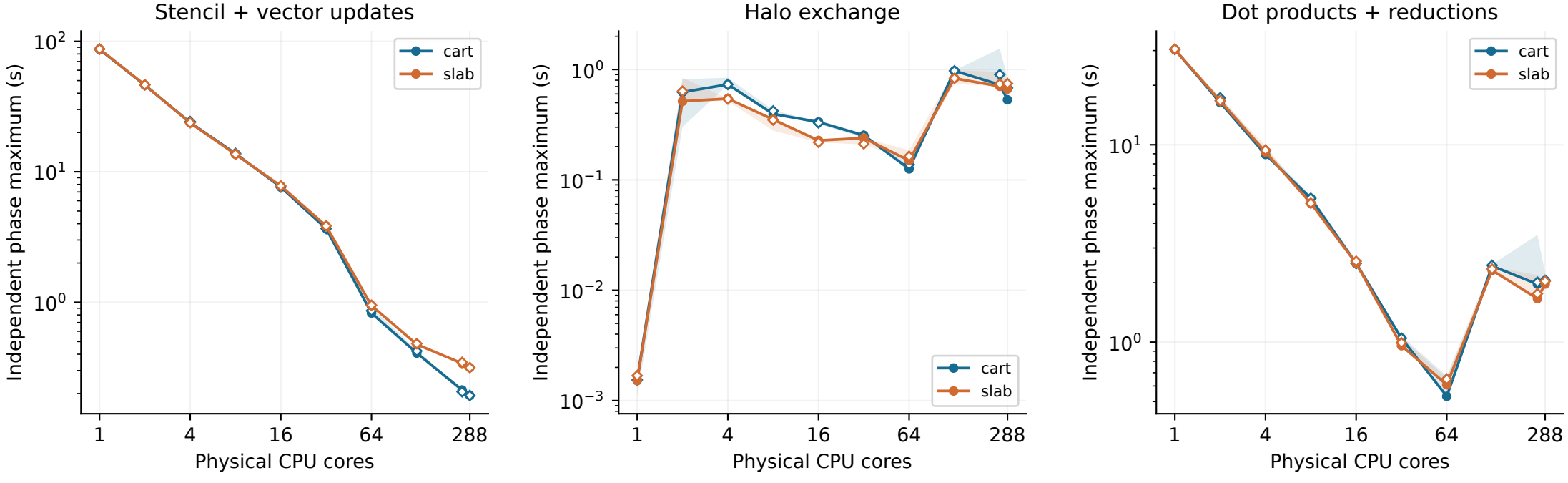




Line/filled marker: minimum; hollow diamond: median; shading: observed min-max (3 runs, not a CI).
Each decomposition uses its own one-rank MPI baseline; ratio shading is an all-pairs range envelope.

# 2D phase timing | n=2048 | independent rank maxima

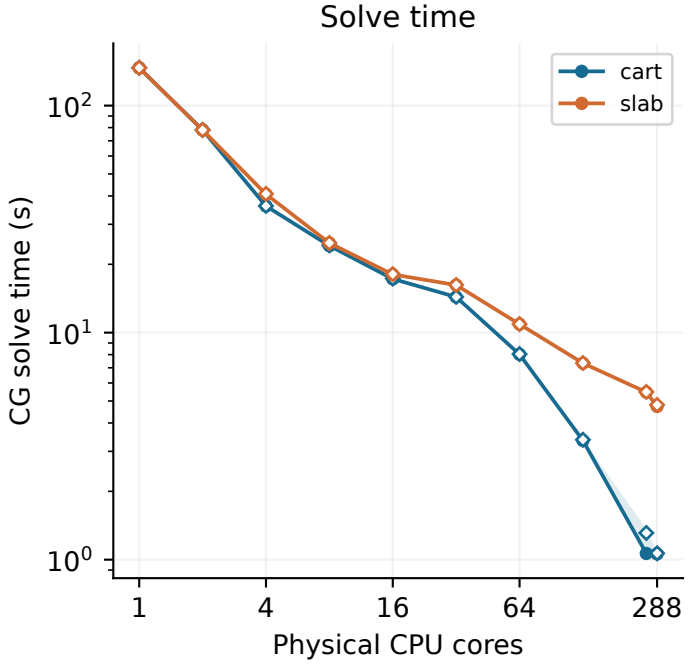
Vanda | audited CPU measurements | 3 independent runs per configuration

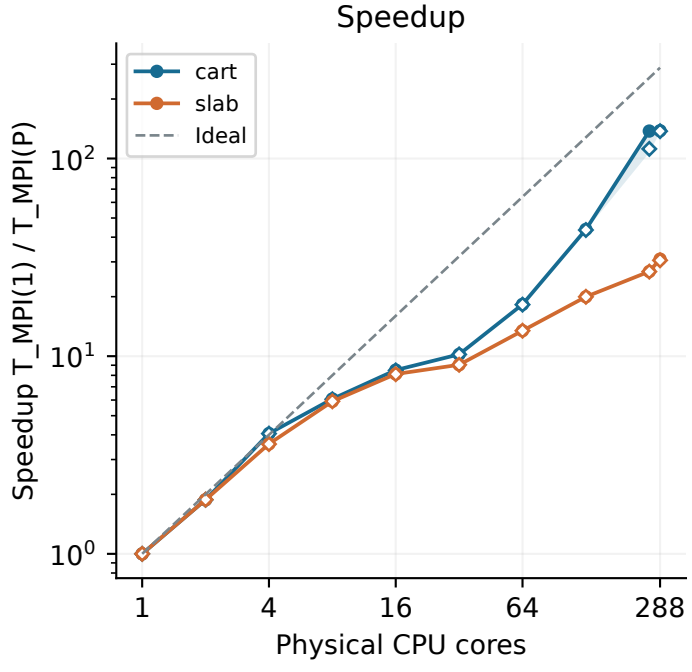


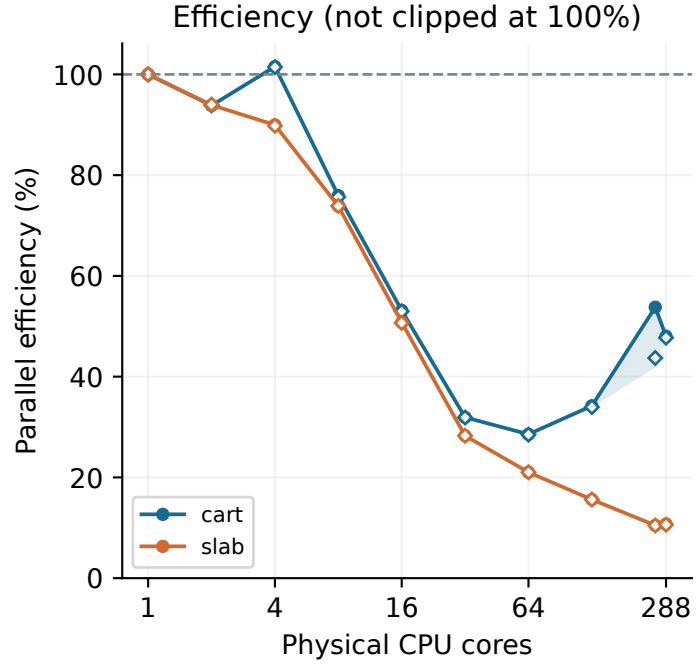

Line: phase from the fastest whole solve; hollow: phase median; shading: observed phase range.
Phase maxima may occur on different ranks and intervals may overlap. Do not stack or interpret their sum as total.

# 3D strong scaling | n=288 | through 288 physical cores

Vanda | audited CPU measurements | 3 independent runs per configuration



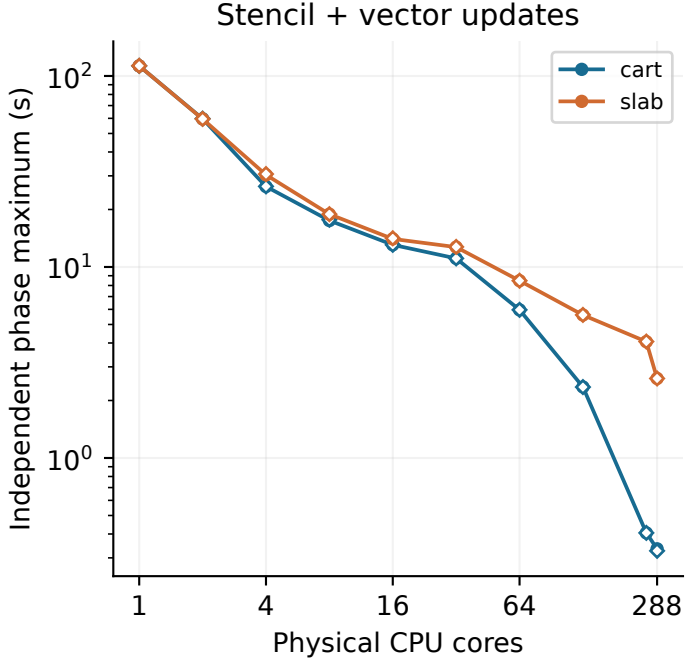



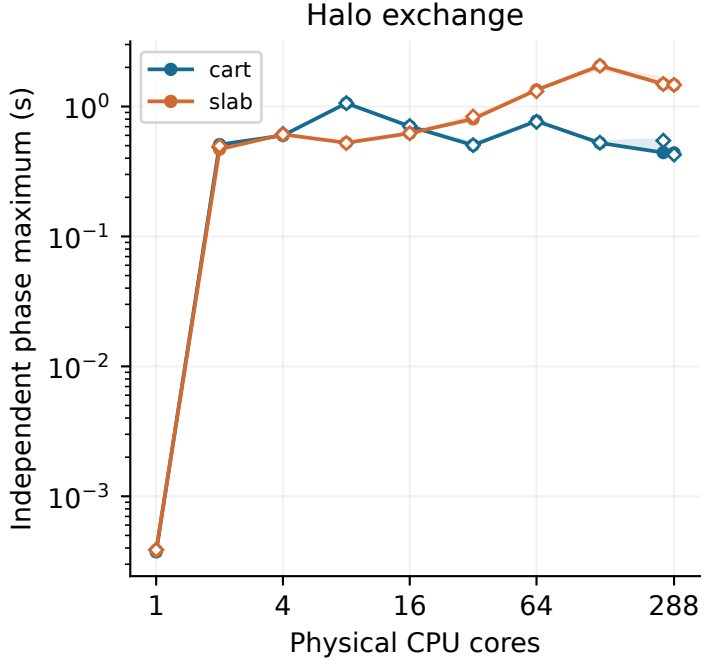



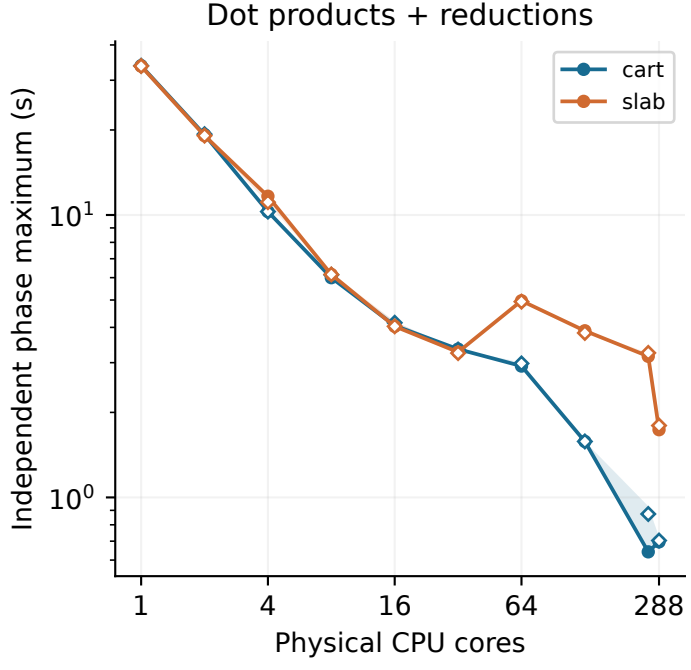

Line/filled marker: minimum; hollow diamond: median; shading: observed min-max (3 runs, not a CI).
Each decomposition uses its own one-rank MPI baseline; ratio shading is an all-pairs range envelope.

# 3D phase timing | n=288 | independent rank maxima

Vanda | audited CPU measurements | 3 independent runs per configuration



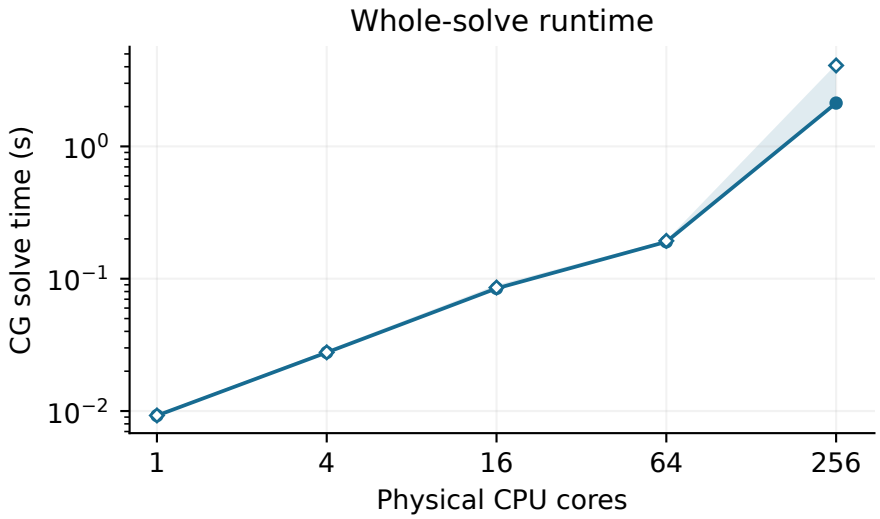



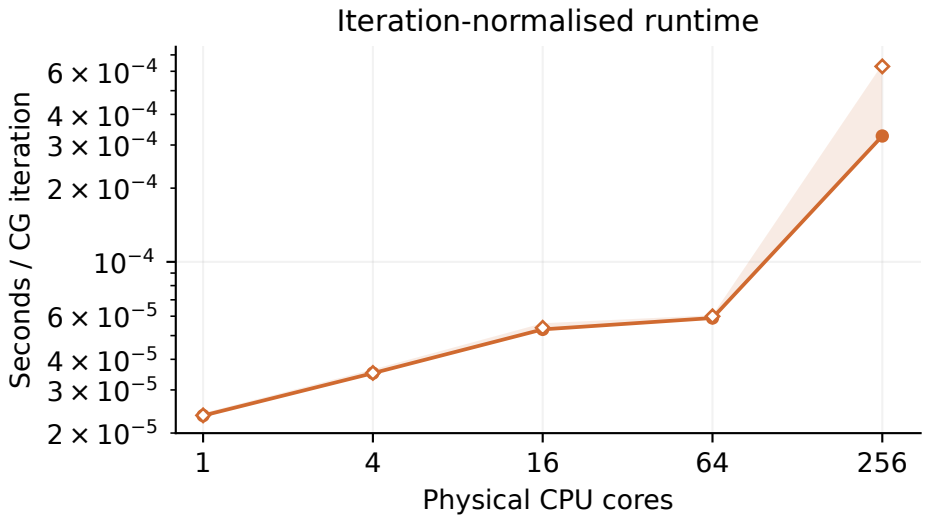



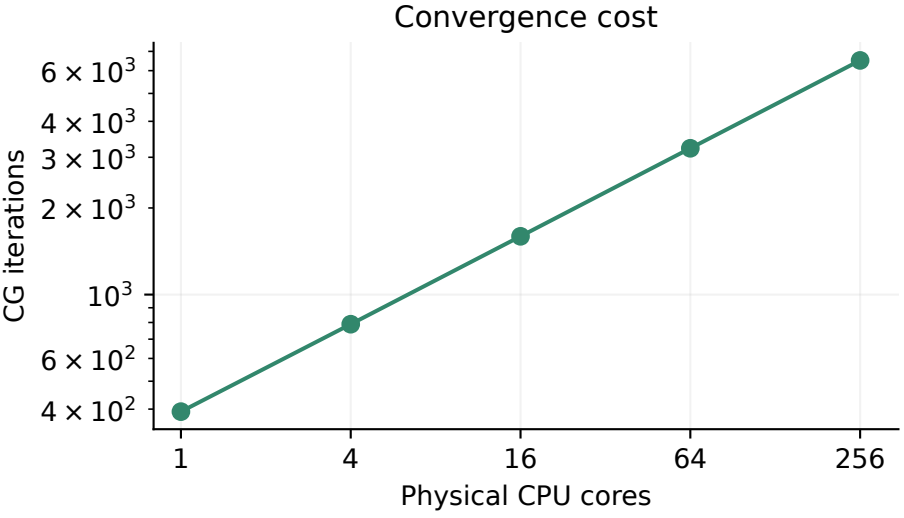

Line: phase from the fastest whole solve; hollow: phase median; shading: observed phase range.
Phase maxima may occur on different ranks and intervals may overlap. Do not stack or interpret their sum as total.

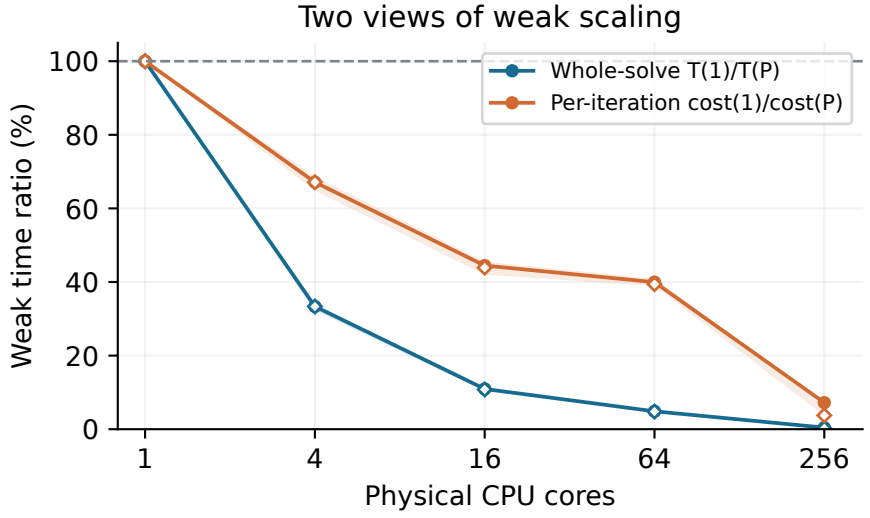
# 2D weak scaling | through 256 physical cores

Vanda | audited CPU measurements | 3 independent runs per configuration



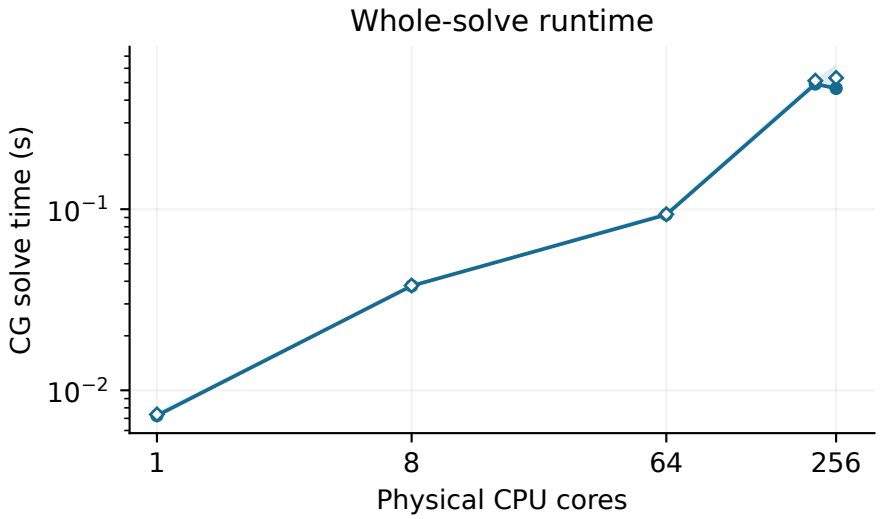



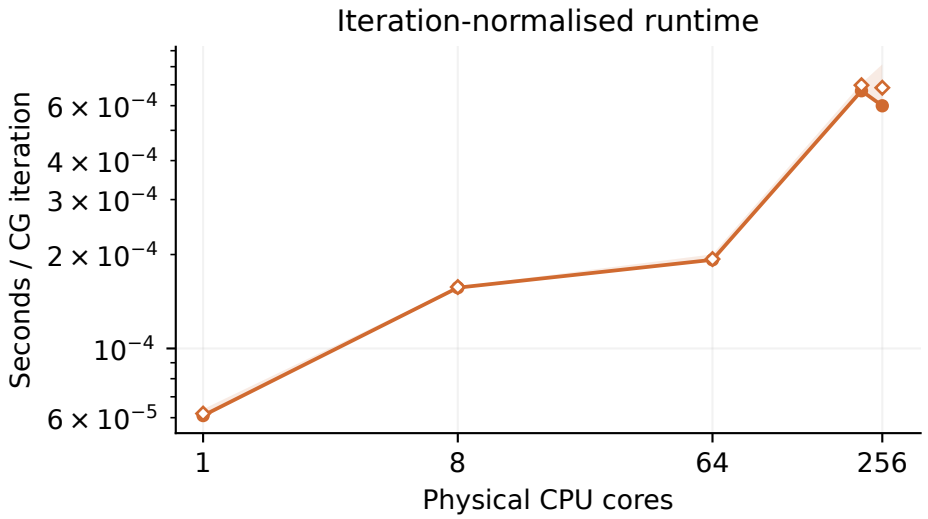



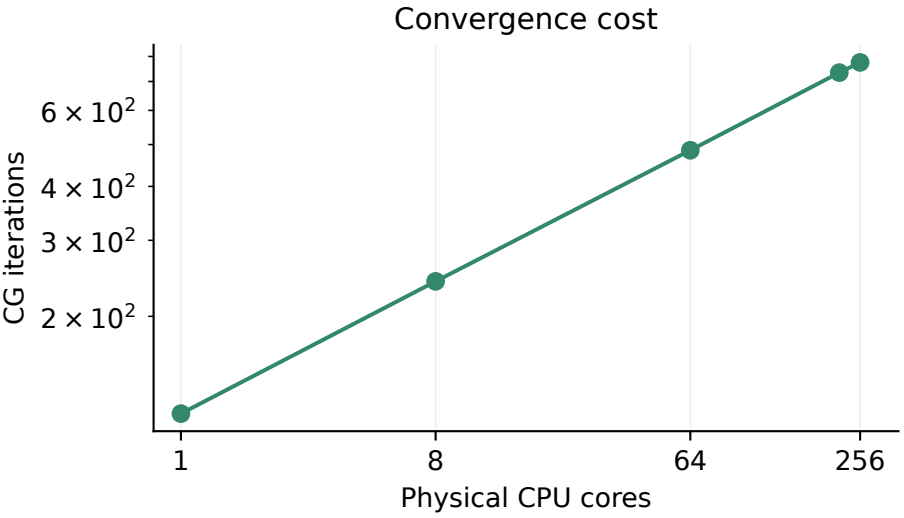

Line/filled marker: minimum; hollow diamond: median; shading: observed min-max (3 runs, not a CI).
2D owned points/core are exactly constant at 16384 through 256 cores.
Whole-solve time includes iteration growth; per-iteration ratios isolate average iteration cost.

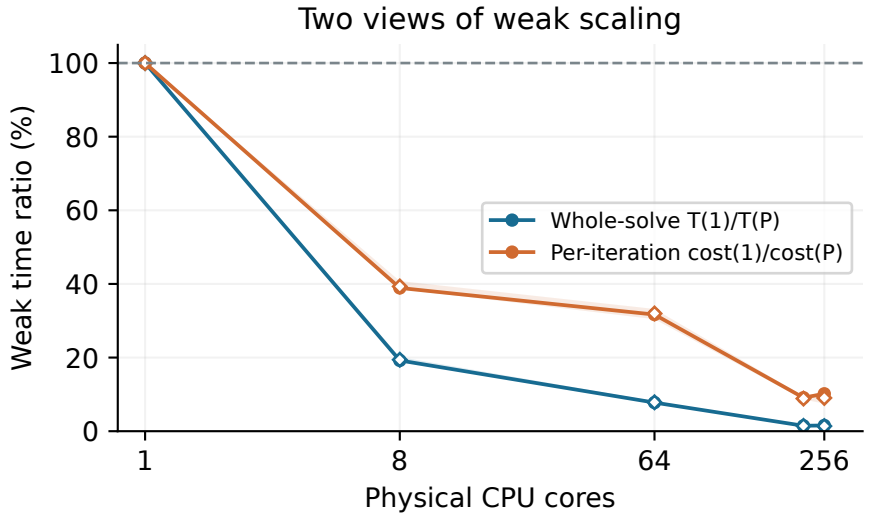
# 3D weak scaling | through 256 physical cores

Vanda | audited CPU measurements | 3 independent runs per configuration



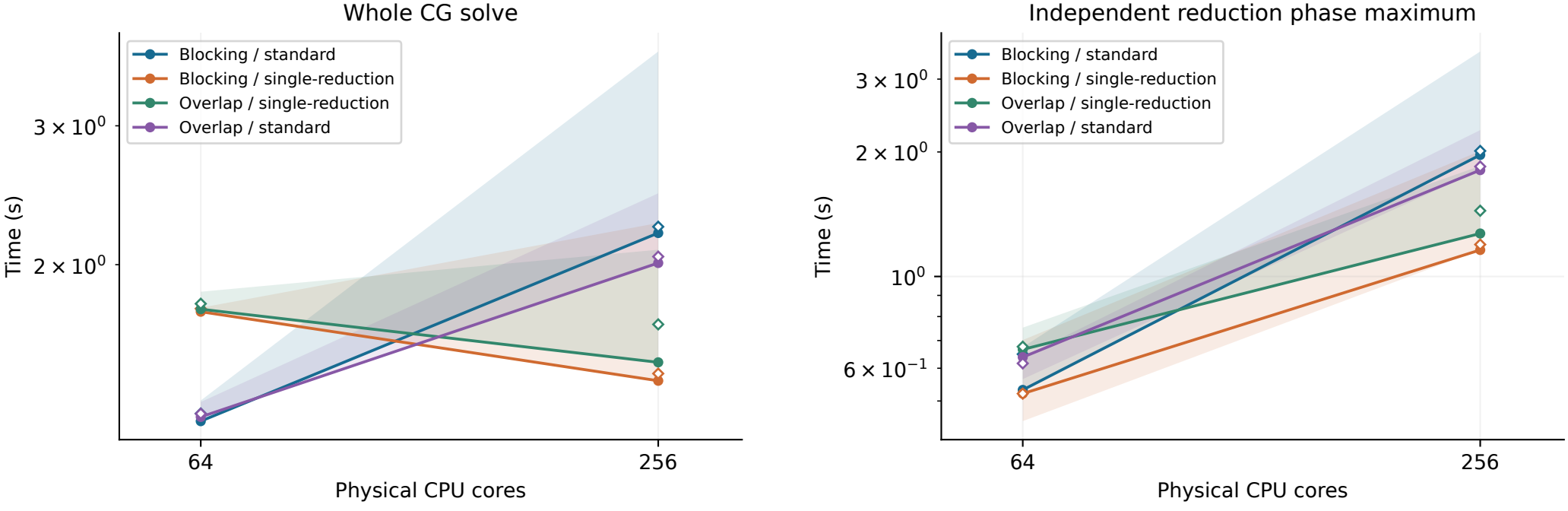




Line/filled marker: minimum; hollow diamond: median; shading: observed min-max (3 runs, not a CI).
3D n=203 at 256 cores gives 32677.449 points/core (-0.276% vs the base).
Whole-solve time includes iteration growth; per-iteration ratios isolate average iteration cost.

# 2D parallelization choices | n=2048 | Cartesian decomposition

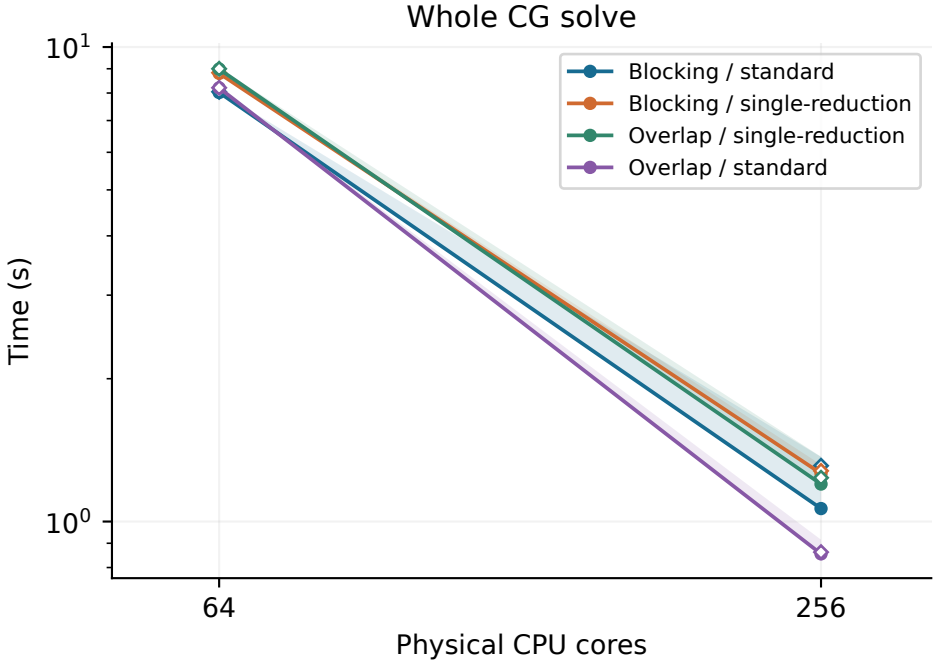
Vanda | audited CPU measurements | 3 independent runs per configuration

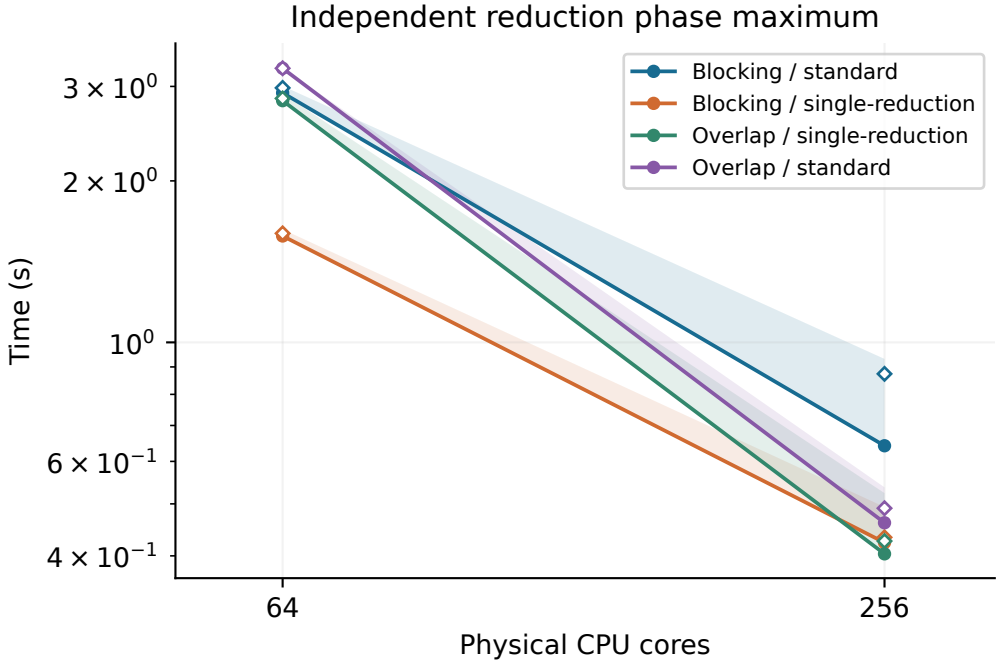



Whole solve: minimum/median/range of 3 runs. Phase line: same row as minimum solve; hollow: phase median.
'pipe' is single-reduction CG (Chronopoulos-Gear); it is not a claim of asynchronous pipelined CG.

# 3D parallelization choices | n=288 | Cartesian decomposition

Vanda | audited CPU measurements | 3 independent runs per configuration

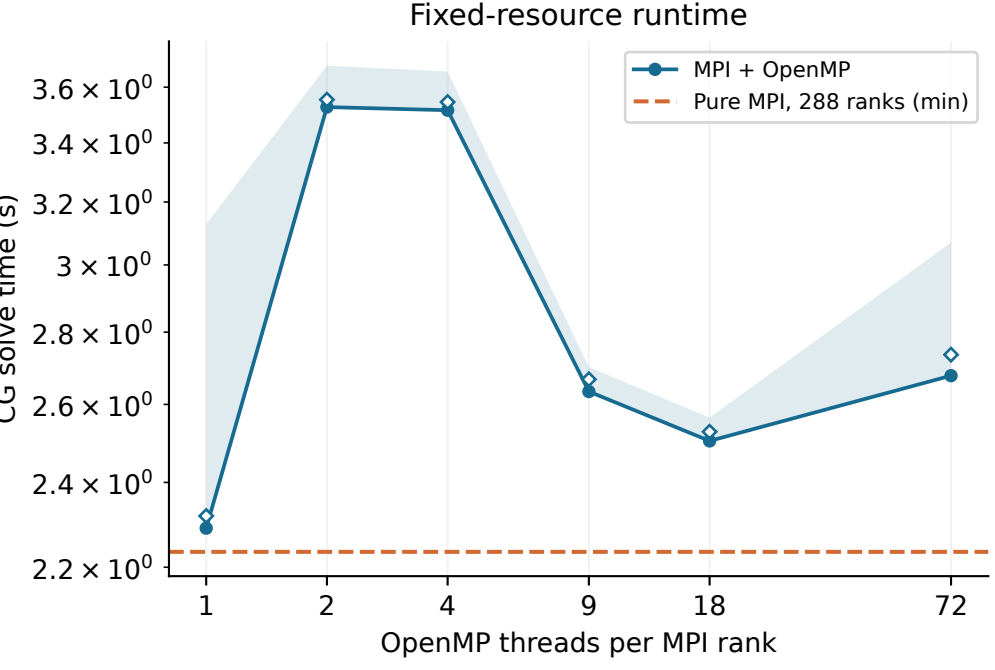


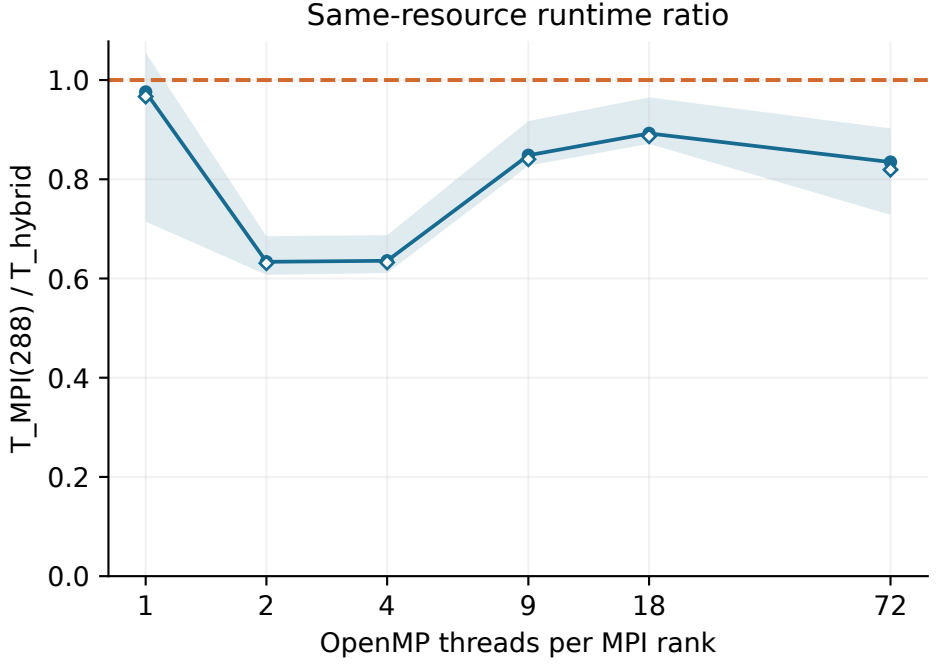




Whole solve: minimum/median/range of 3 runs. Phase line: same row as minimum solve; hollow: phase median.
'pipe' is single-reduction CG (Chronopoulos-Gear); it is not a claim of asynchronous pipelined CG.

# 2D MPI + OpenMP | n=2048 | fixed 288 physical cores

Vanda | audited CPU measurements | 3 independent runs per configuration

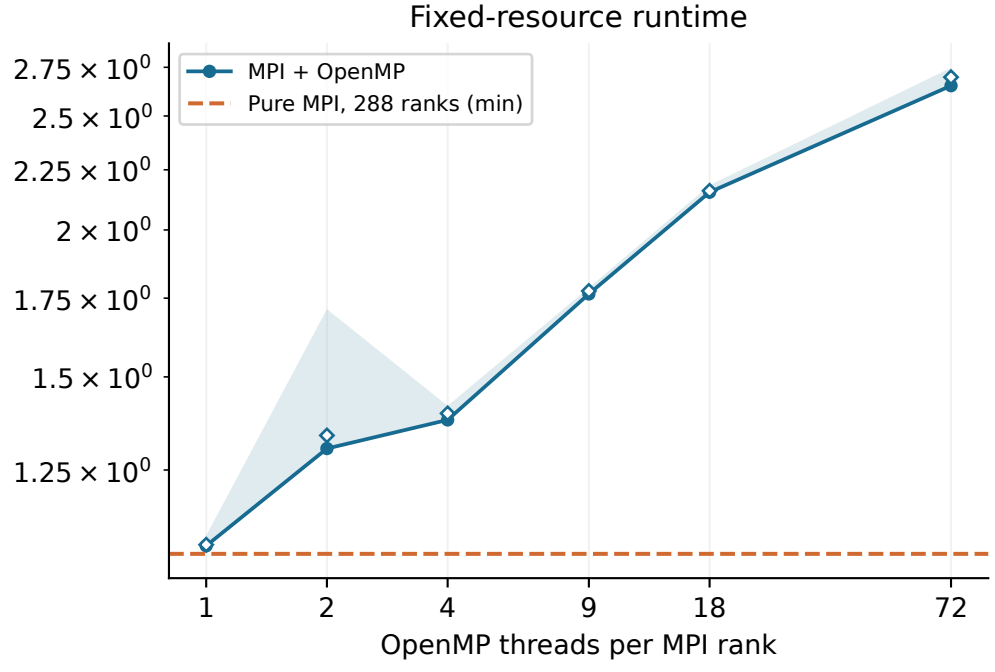


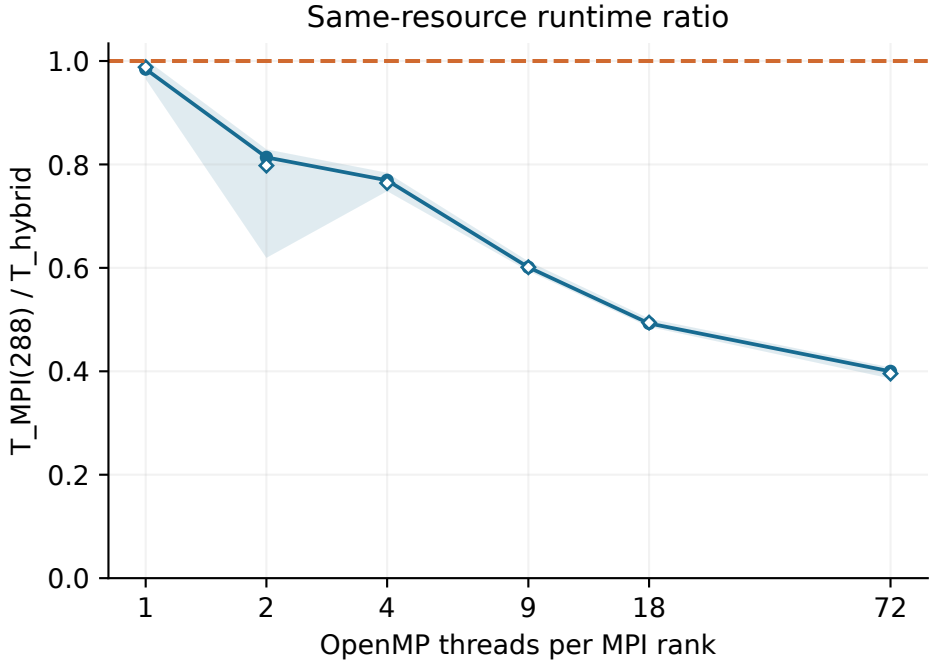




Line/filled marker: minimum; hollow diamond: median; shading: observed min-max (3 runs, not a CI).
MPI ranks x threads: 288x1, 144x2, 72x4, 32x9, 16x18, 4x72. Ratio is not a strong-scaling efficiency.

# 3D MPI + OpenMP | n=288 | fixed 288 physical cores

Vanda | audited CPU measurements | 3 independent runs per configuration




Line/filled marker: minimum; hollow diamond: median; shading: observed min-max (3 runs, not a CI).
MPI ranks x threads: 288x1, 144x2, 72x4, 32x9, 16x18, 4x72. Ratio is not a strong-scaling efficiency.